

Yevhen Biedniakov

Dental CAD/CAM Engineer • Implant & Full-Arch Specialist • AI Automation

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PROFESSIONAL SUMMARY

Dental CAD/CAM engineer with 7 years of digital manufacturing experience, specializing in implant-supported prosthetics, full-arch / All-on-X restorations, and photogrammetry-driven workflows. Production expertise in 3Shape Dental Designer, Exocad, EOSINT M270 laser sintering, and custom implant library configuration across Straumann, ZimVie, Hiossen, BioHorizons, DESS, and Implant Direct. In parallel, build local AI automation systems that compile production data into design-generation pipelines — bridging hands-on lab work with the next generation of dental CAD software.

CORE EXPERTISE

Dental CAD/CAM: 3Shape Dental Designer • Exocad • Meshmixer • Blender • full-arch / All-on-X / All-on-4 • custom abutments • crown & bridge • removable partials • surgical guides • smile design • scan body workflows • Ti-base & screw configuration • library.xml • STL organization

Implant Systems: Straumann • ZimVie / Biomet 3i • Hiossen • BioHorizons • DESS • Implant Direct

Clinical Workflows: photogrammetry alignment • same-day immediate temporaries • chairside & remote surgeon collaboration • digital impression / IOS workflows • passive-fit full-arch verification

Materials & Manufacturing: zirconia • PMMA • e.max • CoCr • titanium • EOSINT M270 laser sintering • 5-axis milling • 3D printing • sintering & finishing protocols

AI / Automation Engineering: Python • RAG pipelines • LLM orchestration • vector search (FAISS) • embeddings • hybrid retrieval (dense + BM25) • reranking • knowledge distillation • XML reverse engineering • SQLite • lxml • Ollama • sentence-transformers • PyTorch • trimesh • PyMeshLab • Docker • Git • Linux

Software Ownership: CAD/CAM configuration • troubleshooting • version management • workflow design • technician training • medical-data & GDPR-aware workflows

PROFESSIONAL EXPERIENCE

CAD/CAM Specialist

Dec 2025 — Present

Unique Metal 3D Printing • Achterveld, Netherlands • On-site

- Design dental frameworks in 3Shape Dental Designer for laser-sintered CoCr production on EOSINT M270.
- Own CAD/CAM software configuration and team training; troubleshoot digital workflow issues across the design-to-print handoff.
- Maintain production-grade output while developing AI automation tools for the same domain in parallel.

Dental CAD/CAM Specialist — Software & Systems

Lintec Dental Labs · Maryland, USA · Hybrid

- Technical owner of the digital department: CAD software configuration, version management, troubleshooting, and lab-wide system updates.
- Designed implant-supported frameworks, bridges, dentures, and full-arch All-on-X prosthetics from planning through delivery.
- Built and customized Exocad implant libraries: STL organization, implant/component mapping, Ti-base and screw configuration, library.xml setup.
- Supported implant systems including Straumann, ZimVie/Biomet 3i, Hiossen, BioHorizons, DESS, and Implant Direct.
- Executed photogrammetry-driven alignment workflows for accurate implant positioning and passive-fit full-arch cases.
- Created real-time CAD designs for post-surgery immediate temporaries, adapting prosthetic geometry to surgical outcomes for same-day delivery.
- Collaborated chairside and remotely with surgeons on same-day cases, syncing CAD adjustments with clinical feedback.
- Trained other technicians on scanning, design protocols, and CAD/CAM troubleshooting.
- Fabricated crowns and bridges, removable dentures, and implant-supported prosthetics while maintaining lab protocols, materials, and equipment.

Web Developer

2016 — 2017

Freelance

- Built responsive web applications using React and TypeScript — foundation for current Python & AI engineering work.

AI & AUTOMATION PROJECTS

Dental Framework AI Design Agent — In Development

Domain-specific AI for CAD/CAM manufacturing · 500GB production dataset

- Multi-stage AI pipeline automating RPD metal framework design using paired production data (intraoral scans + completed CAD designs) from thousands of clinical cases.
- Reverse-engineered proprietary 3Shape XML schema to extract structured design parameters: tooth configurations, materials, insertion vectors, model geometry.
- Hybrid RAG architecture: dense (e5-large-v2) + sparse (BM25), reranking, reciprocal rank fusion.
- Knowledge distillation: cloud teacher models generate synthetic training data; local 2.4GB student model serves offline inference on consumer GPU.
- Generates CAD-ready design parameter files directly — eliminating fragile GUI automation in GDPR-sensitive medical workflows.

Stack: Python · lxml · SQLite · FAISS · Ollama · Gemma 4 · sentence-transformers · trimesh · PyMeshLab

Offline RAG System — Multi-Document Agentic Retrieval

- Fully local RAG pipeline with GPU-powered Ollama inference — no cloud dependencies, no API keys.
- Agentic query decomposition breaks complex questions into sub-queries for stronger cross-document recall.
- Hybrid retrieval with reranking, deduplication, and source attribution by file name and page number.
- Tested across 50+ documents with sub-second retrieval latency on consumer hardware.

Stack: Python · FAISS · sentence-transformers · Ollama · NumPy · PyPDF
github.com/Visiterius/offline-ai-rag-system

Multi-Language Voice AI & Edge Chatbots

- Real-time multilingual voice recognition using Whisper + Claude API; provider-neutral architecture for swappable models.
- Offline LLM chatbots deployed on Raspberry Pi and Pixel 7 with zero internet dependency; quantized-model inference.

Stack: Python · Whisper · Claude API · Ollama · faster-whisper · TypeScript · Android

EDUCATION

Dental Laboratory Technology & Technician — Odesa I.I. Mechnikov National University · 2016–2021

Web Page & Multimedia Design — National University Odessa Law Academy · 2017–2020

JavaScript Algorithms & Data Structures — freeCodeCamp · 2021

LANGUAGES

English — professional · Ukrainian — native · Russian — native · Dutch — A2, targeting B1/B2